

# Joseph Walton

Senior AI Engineer

Email: [jwgtwalton@gmail.com](mailto:jwgtwalton@gmail.com)

Telephone: 07817685781

LinkedIn: [linkedin.com/in/joseph-walton-2a508591](https://www.linkedin.com/in/joseph-walton-2a508591)

GitHub: [github.com/jwgtwalton](https://github.com/jwgtwalton)

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## Professional Summary

Senior AI Engineer with 10+ years' experience delivering production AI solutions across regulated government, financial services, intellectual property and defence environments. Hands-on expertise in Generative AI, RAG, agentic AI, Azure OpenAI, Microsoft AI Foundry, LangChain and LangGraph, with strong Python and FastAPI development experience. Experienced in prompt engineering, LLM evaluation and observability, MLOps, secure deployment and enterprise integration. Proven technical leader with production solutions delivering £2M+ annual savings and significant improvements in performance and cost.

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## Technical Skills

- **Generative AI:** Azure OpenAI, Microsoft AI Foundry, LangChain, LangGraph, RAG, agentic AI, prompt engineering, LLM evaluation, observability, MCP
  - **Engineering:** Python, FastAPI, REST APIs, SQL, Java, microservices, Azure Functions
  - **Azure & Data:** Azure AI Search, Cognitive Services, Azure Databricks, Apache Spark, Elasticsearch, vector search, HNSW, Polars
  - **MLOps:** Azure ML Pipelines, Vertex AI Pipelines, MLflow, Docker, CI/CD, model evaluation and deployment
  - **Security:** Secure by Design, AI red teaming, adversarial testing, secure containerised deployment, responsible AI
  - **Machine Learning:** Hugging Face, TensorFlow, PyTorch, Scikit-learn
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## Professional Experience

### IBM

Lead Data Scientist

June 2025 – Present | Contract – Central Government

- Designed and delivered a production agentic AI solution using Microsoft AI Foundry, LangGraph and Azure Functions, saving case workers approximately two hours per day and delivering ~£2M annual savings.
- Built an MCP server exposing a data-matching algorithm for reuse across government AI applications.
- Developed prompt engineering, LLM evaluation and observability practices to improve the reliability and performance of production AI applications.
- Built human-in-the-loop LLM evaluation workflows, reducing evaluation effort by approximately one hour per case.
- Re-engineered a data-matching algorithm using Polars, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k/month.

- Worked with architecture, engineering and business stakeholders to deliver secure, maintainable production AI solutions.

**Tech Stack:** Python, Microsoft AI Foundry, Azure OpenAI, LangGraph, MCP, Azure Functions, Polars

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## Intellectual Property Office

### Senior Machine Learning Engineer

*November 2021 – April 2025*

- Developed production Generative AI applications using Python, LangChain, Azure OpenAI and Azure Functions.
- Built a RAG agent for identifying relevant Goods and Services terms, integrating LLM reasoning with enterprise search and domain-specific data.
- Implemented and optimised prompt engineering and LLM evaluation, improving application reliability and response quality.
- Tuned Elasticsearch HNSW vector search, improving search and recommendation relevance by 20% while balancing throughput and recall.
- Implemented end-to-end MLOps and CI/CD across multiple environments, reducing model deployment time to under one day.
- Conducted Generative AI red-team exercises and advised technical and non-technical stakeholders on secure and responsible AI adoption.

**Tech Stack:** Python, Azure OpenAI, LangChain, Azure Functions, Azure AI Search, Azure ML Pipelines, Elasticsearch, Azure Databricks

## Earlier Experience

### Incopro

#### Lead Machine Learning Engineer / Machine Learning Engineer

*November 2018 – November 2021*

- Led delivery of secure production ML systems.
- Developed and deployed scalable Python/FastAPI services supporting machine learning and data-processing workloads.
- Designed an end-to-end MLOps framework using Vertex AI Pipelines, orchestrating training, evaluation, approval and deployment.
- Developed multilingual NLP models covering 90+ languages and built scalable text, image and recommendation systems.
- Mentored engineers transitioning into machine learning roles.

**Tech Stack:** Python, FastAPI, Hugging Face, TensorFlow, PySpark, Vertex AI Pipelines, MLflow, Docker, GCP

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## Office for National Statistics

### Machine Learning Engineer

*September 2017 – November 2018*

- Developed Python ML services and scalable data pipelines processing sensitive administrative datasets.
  - Contributed to the backend architecture for the UK Census 2021 data-collection system.
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## Barclays

**Data Scientist — Secondment**

*September 2018 – October 2018*

- Applied statistical and analytical methods to securely use payments data for regional economic indicators.
  - Member of the winning ONS × Barclays Data Hackathon team.
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## Purple Secure Systems

**Software Engineer**

*August 2014 – October 2016*

- Developed Python and Java big-data processing systems for government and defence clients.
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## Education

### University of Bath

- **MSc Applied Mathematics** — 2016–2017
  - **BSc Physics** — 2010–2013
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## Award

### Winner — ONS × Barclays Data Hackathon

Used payments data to develop more granular and timely estimates of Gross Value Added (GVA).